

PHY201 : Waves and Optics
First Mid-Semester Examination 2025
Duration : 1 hour

Instructor: A. Chaudhuri

Max. Marks 20

- All questions are compulsory.
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1. A mass–spring system with mass $m = 1.00$ kg and spring constant $k = 100$ N m⁻¹ is released at $t = 0$ with $x(0) = 0.030$ m and $\dot{x}(0) = -0.40$ m s⁻¹.
 - (a) Write the equation of motion and determine the natural angular frequency ω_0 . [2]
 - (b) Determine A and ϕ from the initial conditions and express the motion as $x(t) = A \cos(\omega_0 t + \phi)$. [3]
 - (c) Compute the total mechanical energy E . [1]
2. A lightly damped oscillator satisfies $\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$ with $\omega_0 = 20\pi$ s⁻¹. Experimentally, the *amplitude* drops by a factor of e in $N = 40$ cycles.
 - (a) Find the damping constant γ . [2]
 - (b) Estimate the quality factor Q . [1]
 - (c) Write the damped angular frequency ω and comment (one line) on the shift relative to ω_0 . [1]
3. A lightly damped oscillator has quality factor $Q = 50$ and frequency $f = 5$ Hz.
 - (a) Using the small-damping relation, find the *fractional energy loss per cycle*. *Hint:* Start from $E(t) = E_0 \exp(-\gamma t)$. [2]
 - (b) Determine the *amplitude ratio* A_N/A_0 after $N = 25$ cycles. [2]
4. Consider a damped oscillator with $m = 0.25$ kg, $b = 2$ N s m⁻¹, $k = 25$ N m⁻¹, driven by

$$F(t) = F_0 \cos(\omega t), \quad F_0 = 10 \text{ N}, \quad \omega = 10 \text{ s}^{-1}.$$

In the steady state, the response is $x(t) = A \cos(\omega t - \delta)$.

- (a) Obtain A and δ . (*Give explicit formulas in terms of m, k, b, ω, F_0 and then evaluate numerically.*) [2]
- (b) Compute the *energy dissipated per cycle* against the resistive force. [2]
- (c) Find the mean power input. [2]